TT-I-540C
June 18, 1973
SUPERSEDING
Int. Fed. Spec. TT-I-00540b(GSA-FSS)
February 10, 1967 and
Fed. Spec. TT-I-540
December 15, 1955

FEDERAL SPECIFICATION

INK, NUMBERING AND DATING MACHINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ink suitable for use with hand- and electrically operated numbering and dating machines, as applicable, for general office use. The ink is furnished in 2-ounce containers, unless otherwise specified (see 3.7).

1.2 Classification.

1.2.1 Type and grade. The numbering and dating machine ink shall be of one type and grade.

1.2.2 Colors. (See 3.6.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

GG-N-760 - Numbering and Dating Machines, Office Type, Electric.
GG-N-775 - Numbering Machines, Office Type, Hand-Operated.
UU-P-121 - Paper, Bond and Writing, White and Colored.
UU-P-328 - Paper, Manifold.
UU-P-547 - Paper, Teletypewriter, Roll.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, a Handbooks as outlined under General Information in the Index of Federal Specifications and Standards a at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issu is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Offic Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbook and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Stteet, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The manufacturer is given wide latitude in the selection of materials and processes of manufacture provided the manufactured ink meets the requirements of this specification.

3.2 Stability. There shall be very little evidence of separation and no evidence of caking of the coloring matter in the ink (see 4.4.1).

3.3 Performance. The ink shall be in a smooth working condition. Inked impressions shall present clear and legible characters of uniform intensity without evidence of feathering when tested as specified in 4.4.2.

3.4 Drying time. Inked impressions shall dry to the extent that the ink will not blur or smear when tested in accordance with 4.4.3.

3.5 Fading. Inked impressions shall be clear and readily legible and shall not show a degree of fading which will interfere with the legibility of the inked impressions when tested as specified in 4.4.4.

3.6 Colors. Colors of the ink shall be black, blue, and red, as specified (see 6.2).

3.7 Unit containers (bottles). Unless otherwise specified (see 6.2), the ink shall be furnished in 2-ounce containers (see 5.1.1.1 and 5.1.2.1).

3.8 Identification marking (bottles). Unless otherwise specified, each bottle of ink shall be clearly and legibly marked or labeled with the following information.

Ink, Numbering and Dating Machine
Fed. Spec. TT-I-540 (latest issue)
Color
Quantity of ink
Fed. Stock number
Contract number
Manufacturer's name or trademark

Additional marking in accordance with the manufacturer's normal commercial practice will be acceptable. Unless otherwise specified, marking ink shall be water resistant and labels shall be printed with water-resistant ink on a good grade of printing paper firmly attached to the bottle with a water-insoluble adhesive. (See 6.2.)

3.9 Workmanship. The ink shall be smooth, homogeneous, and free from lumps and strings. Labels, when used, shall be neatly and securely attached to individual bottles. Bottles shall be clean and there shall be no evidence of spillage or leakage of ink. Workmanship shall be in accordance with good commercial practice.

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3.13 Shelf life. The ink in original, unopened individual containers and not subjected to extremes of temperature and humidity shall retain its utility and serviceability as defined by this specification for not less than one year from date of receipt by the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.2).

4.2 Sampling and inspection for acceptance. Sampling and inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Sampling.

4.2.1.1 Lot. For the purpose of sampling, an inspection lot offered for acceptance inspection and tests shall consist of all material and all containers (bottles) of ink submitted for inspection and acceptance at one time.

4.2.1.2 Sampling for visual inspection of filled containers. (See 4.2.2.2.1.1 and 4.2.2.2.1.2.)

4.2.1.3 Sampling for lot acceptance tests. One sample unit shall consist of not less than 2 fluid ounces of ink, in the original unopened container and bearing all the manufacturer's identification markings. The sample unit shall be taken at random from each inspection lot (4.2.1.1).

4.2.2 Inspection.

4.2.2.1 Inspection of materials and components. The supplier is responsible, in accordance with 4. for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specific requirements of referenced subsidiary specifications and standards and in accordance with this specification, as applicable. In case of conflict, the requirements of this specification sh govern.

4.2.2.2 Inspection of end item and preparation for delivery.

4.2.2.2.1 Examination of end item. The end item shall be examined in accordance with the defects listed for each of the applicable examinations.

4.2.2.2.1.1 Examination of individual containers of ink. An examination shall be made for defects as indicated in table I. The sample unit for this examination shall be one 2-ounce glass or polyethyle bottle of ink, as applicable, taken from the lot offered for acceptance as specified in 4.2.1.1. The inspection and acceptable quality levels (AQL's) shall be as specified in 4.2.3 with the lot size expressed in units of individual containers of ink.

(table I, see page 4)

4.2.2.2.1.2 Examination of individual containers for average contents. The sample unit for this examination shall be one 2-ounce glass or polyethylene container, as applicable, taken from the lot offered for acceptance as specified in 4.2.1.1. The inspection levels and AQL's shall be as specified in 4.2.3 with the lot size expressed in units of individual containers of ink. The average contents shall be not less than that specified.

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TABLE I. Visual examination (end item)

Examine	Defects
Appearance	Spillage of ink or leakage on outside of bottle. Bottles not clean. Labels damaged, torn; not securely attached to bottle.
Design	Container and closure not as specified; cap w/o brush.
Marking (bottle)	Not clear nor legible; not as specified.
Fill of individual container (except average contents)	Empty or broken bottle; fill not as indicated.
Workmanship	Not as specified.

4.2.2.2.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5 and the contract or order. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container prepared for delivery and selected just prior to the closing operations. Defects of closure specified shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end-item inspection lot. The inspection level shall be S-2, with an AQL of 4.0 defects per 100 units.

TABLE II. Examination of preparation for delivery

Examine	Defects
Markings - (interior and exterior containers, as applicable)	Incorrect; omitted; illegible; improper size, location, sequence, or method of application; ommission "THIS SIDE UP" and other warnings, as applicable.
Materials	Wrong type, class, or other category of required specifications. Material missing, damaged, or defective affecting serviceability.
Workmanship	Wrong or careless means in placement and use of labeling and packaging and packing materials such as loose or partially attached labels; flaps of containers incompletely closed; loose or over-tight straps, dangerous strap ends, protruding nails, and smeared adhesive.
Contents (interior and exterior containers, as applicable)	Quantities of unit containers not as specified. Wrong arrangement; bottle on side or upside down.
Closure of shipping containers	Not as specified.
Gross weight of shipping container	In excess of specified weight. In excess of box limitation shown on boxmaker's certificate.

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4.2.3 Inspection levels and AQL's. The inspection levels and AQL's for examination of the end item shall be as follows:

Specification reference	Inspection level	AQL
4.2.2.2.1.1	S-4	2.5
4.2.2.2.1.2	S-2 (See 4.2.2.2.2)	---

4.3 Lot acceptance tests.

4.3.1 Testing of ink for lot acceptance. The sample ink taken in accordance with 4.2.1.3 shall be subjected to tests outlined in 4.4. Failure of the ink to meet the requirements of any one or more of these tests shall be cause for rejection of the entire batch or lot offered for delivery.

4.4 Tests.

4.4.1 Stability. Shake the ink container vigorously for 10 seconds. Transfer 50 milliliters of the ink into a glass test tube. After a period of 24 hours, examine the contents for separation and caking.

(Note. - Any separation observed should be easily returned to a homogeneous suspension after agitatic for 10 seconds.)

4.4.2 Performance. This test shall be performed with a hand-operated, self-inking numbering machin conforming to GG-N-775 and equipped with a new unused inking pad. Ink the numbering machine pad evenly with the sample ink, applying a sufficient quantity to produce full colored impressions. (Caution: To much ink will cause slow drying, bleeding, and uneven impressions.) On a flat metal or other rigid, level surface, place a sheet of 32-pound writing paper conforming to type V of UU-P-121, evenly upon tw sheets of 18-pound manifold paper conforming to type V of UU-P-328. Place the numbering machine on the writing paper and proceed to make a series of 25 impressions using the same number. Then make another series of 25 impressions by advancing the numbers in regular progression. After a 10-minute waiting period, examine the impressions for legibility, intensity, uniformity, and feathering. Repeat the test on the first and second copies of a nine part surface coated paper conforming to type 1, class 5, of UU-P-547.

4.4.3 Drying time. Make an inked impression on the writing paper and the first and second copies of a nine part surface coated paper conforming to type 1, class 5, of UU-P-547 with the hand-operated numbering machine in the usual way. After 1 minute, rub the impressions lightly with the thumb or forefinger and examine for smearing or blurring.

4.4.4 Fading. One-half of the sheet on which the impressions have been made, as described in 4.4.2 shall be exposed to the radiation from a glass-enclosed carbon arc lamp such as the Fade-Ometer, type FDA-S, or its equivalent. Blue and black ink shall be exposed for 24 hours; red ink, 12 hours. The exposed sample shall show not more than the degree of fading as specified in 3.5 when compared with the unexposed sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Ink, furnished in 2-ounce quantities (see 3.7), shall be packaged in a gl or polyethylene bottle provided with a screw-cap and attached brush.

5.1.1.2 Intermediate packaging. Twelve filled bottles of ink of like description shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Top and bottom fiberboard or polystyrene pads, the full size of the inside of the box, shall be provided. Eac bottle shall be inserted in an individual full-height cell formed by slotted polystyrene or fiberboard

5.1.2 Level B.

5.1.2.1 Unit packaging. Ink, furnished in 2-ounce quantities (see 3.7), shall be packaged in a glass or polyethylene bottle provided with a screw-cap and attached brush.

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5.1.2.2 Intermediate packaging. Twelve filled bottles of ink of like description shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Top and bottom fiberboard or polystyrene pads the full size of the inside of the box shall be provided. Each bottle shall be inserted in an individual full-height cell formed by slotted fiberboard or polystyrene.

5.1.3 Level C. The ink shall be packaged to afford protection against leakage, deterioration, and damage in shipment from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages of ink of like description shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve intermediate packages of ink of like description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with Method II of the appendix to the box specification.

5.2.3 Level C. The ink shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking. (See 3.8.) In addition to the special marking required in 5.3.3, all marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to markings required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with Fed. Std. No. 123, including specification classification (such as color, size, etc.).

5.3.2 Military agencies. In addition to markings required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. Each shipping container containing glass bottles shall bear "FRAGILE" markings. All shipping containers shall bear arrows and the word "UP" on at least one side and one end and the words "TOP" and "THIS SIDE UP" on the top surface.

6. NOTES

6.1 Intended use. Numbering machine ink covered by this specification is intended for use with numbering and dating machines as covered in Federal Specification GG-N-760, Numbering and Dating Machines, Office-Type, Electric; and GG-N-775, Numbering Machines, Office Type, Hand-Operated.

6.2 Ordering data. Purchasers should select the preferred options offered herein (see 3.7 and 3.8) and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color of ink required (see 3.6).
(c) Data; inspection responsibility if other than as specified herein (see 4.1).
(d) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).
(e) Whether civil or military marking is required (see 5.3).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired commodity and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

COORDINATION WAIVED BY DOD

User Interest	Preparing Activity:
Army - GL	GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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